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Continuous
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Expected value
and variance

Special
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distribution

Lecture notes for Applies statistics, Probability theory

Lecture 4.

Dr. Barta Attila¹, Dr. Pecsora Sándor²

^{1,2}University of Debrecen, Faculty of informatics,
Department of Applied statistics and probability theory

CS BSc, CSE BSc



Presentation Overview

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Random variables



Definition

Given a sample space Ω , a set of events $\mathcal{F}$, a probability function $\mathbb{P}$, and the set of real numbers $\mathbb{R}$, a random variable X is a function with domain Ω and range $\mathbb{R}$.

$$X : \Omega \rightarrow \mathbb{R}$$

Example

We flip a coin. Whenever the coin comes up heads, we report 1; when it comes up tails, we report 0. This is a random variable.

$$X(\{\text{HEAD}\}) = 1 \quad X(\{\text{TAIL}\}) = 0$$



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Cumulative distribution function, CDF

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Definition

The cumulative distribution function (**CDF**) of a real-valued random variable X is the function given by:

$$F_X(x) = P(X < x)$$

where the right-hand side represents the probability that the random variable X takes on a value less than x .

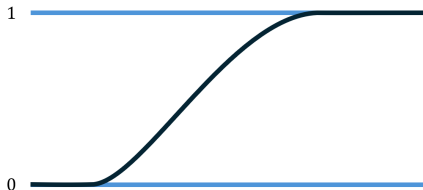


Fig. 1: CDF of continuous random variable



Cumulative distribution function, CDF

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Remarks

- ① Any random variable has a CDF.
- ② The CDF is a real function.
- ③ To calculate $F(t)$ first we should fix the value of t and then find the value of $F_X(x) = P(X < x)$.

Properties of the CDF

- ① F is monotone increasing.
- ② F is left continuous.
- ③ $\lim_{t \rightarrow -\infty} F(t) = 0$ and $\lim_{t \rightarrow \infty} F(t) = 1$



Definition

Let F be the CDF of X . We say that the distribution of X is absolutely continuous, if there exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ so that

$$F(t) = \int_{-\infty}^t f(s) ds, \forall t \in \mathbb{R},$$

f is called the probability density function (PDF) of X .

Remark

- ① Usually we use $f(t) = F'(t)$
- ② If the PDF exists, then the CDF should be continuous. Therefore a discrete random variable has no PDF.



The properties of the PDF

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Properties

$f : \mathbb{R} \rightarrow \mathbb{R}$ is PDF

- ① $\lim_{x \rightarrow -\infty} f(x) = 0,$
- ② $\lim_{x \rightarrow \infty} f(x) = 0,$
- ③ $\int_{-\infty}^{\infty} f(x) dx = 1.$



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Expected value

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The expectation of a random variable is its mean value or its average. In the discrete case it was

$$\mathbb{E}X = \sum_i x_i P(X = x_i)$$

We can not apply it for absolutely continuous distributions because in that case

$$P(X = x_i) = 0, \forall x_i \in \mathbb{R}.$$

However, if we divide the real line into short intervals, then we obtain a similar formula

$$\mathbb{E}X \approx \sum_i x_i P(x_i \leq X < x_{i+1}) = \sum_i x_i [F_X(x_{i+1}) - F_X(x_i)],$$

where F_X is the CDF. If X has PDF f_X , then

$$\mathbb{E}X \approx \sum_i x_i \int_{x_i}^{x_{i+1}} f_X(x) dx \approx \int_{-\infty}^{+\infty} x f_X(x) dx.$$



Definition

Let the random variable X have PDF f . If $\int_{-\infty}^{+\infty} |x|f(x)dx$ is finite, then we say that X has a finite expectation. In this case the quantity

$$\mathbb{E}X \int_{-\infty}^{+\infty} xf_X(x)dx.$$

exists, it is finite, and it is unique. Then the number $\mathbb{E}X$ is called the expectation of X .



Theorem

Assume that the expectation of X is finite. Let $c \in \mathbb{R}$. Then the expectation of cX is finite and

$$\mathbb{E}(cX) = c\mathbb{E}X.$$

If the expectations of X and Y are finite, then the expectation of $X + Y$ is also finite and

$$\mathbb{E}(X + Y) = \mathbb{E}X + \mathbb{E}Y.$$



Theorem

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Theorem

- ① If $X \geq 0$, then $\mathbb{E}X \geq 0$.
- ② If $X \geq Y$, then $\mathbb{E}X \geq \mathbb{E}Y$.
- ③ If $X \geq 0$ and $\mathbb{E}X = 0$, then $P(X = 0) = 1$.



Moments

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Let $k \geq 0$. The k^{th} moment of X is

$$\mathbb{E}X^k.$$

It can be calculated as

$$\mathbb{E}X^k = \int_{-\infty}^{\infty} x^k f(x) dx,$$

where f is the PDF of X .



Definition

The variance of X is

$$\text{Var}(X) = \mathbb{E}(X - \mathbb{E}X)^2 = \mathbb{E}X^2 - \mathbb{E}^2X.$$

Proposition

$$\text{Var}(aX + b) = a^2 \text{Var}(X).$$

If X and Y are independent, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$



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Uniform distribution

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Choose a point randomly from the interval $[a, b]$. Denote by X the position of the point. We say that X has uniform distribution on the interval $[a, b]$. The CDF of X is

$$F(t) = \begin{cases} 0, & \text{if } t \leq a, \\ \frac{t-a}{b-a} & \text{if } a < t \leq b, \\ 1, & \text{if } b < t. \end{cases}$$



The CDF of the uniform distribution

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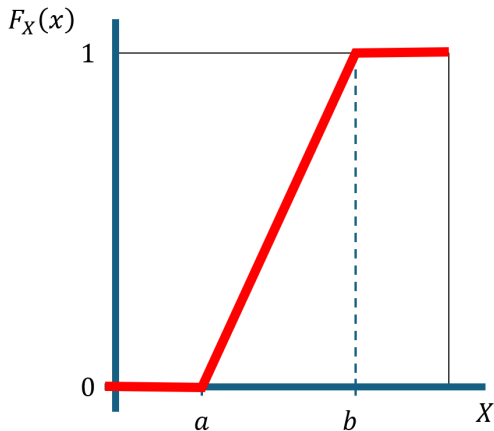


Fig. 2: The CDF of the uniform distribution



The PDF of the uniform distribution

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$$f(t) = \begin{cases} \frac{1}{b-a}, & \text{if } t \in [a, b], \\ 0 & \text{if } t \notin [a, b]. \end{cases}$$

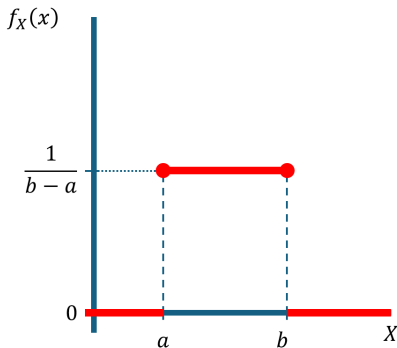


Fig. 3: The PDF of the uniform distribution



The expectation and the variance of the uniform distribution

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The expectation is:

$$\mathbb{E} = \frac{a+b}{2}.$$

The second moment is:

$$\mathbb{E}^2 = \int_a^b \frac{x^2}{b-a} = \frac{b^3 - a^3}{3(b-a)} :$$

So the variance is:

$$\text{Var}(X) = \mathbb{E}X^2 - \mathbb{E}^2X = \frac{b^2 + ab + a^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12}$$



Normal distribution

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The simplest case of a normal distribution is known as the standard normal distribution or unit normal distribution. This is a special case when $\mu = 0$ and $\sigma^2 = 1$.

Every normal distribution is a version of the standard normal distribution, whose domain has been stretched by a factor σ and then translated by μ .

The CDF of the Standard normal distribution is:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt$$



The CDF of the Standard Normal distribution

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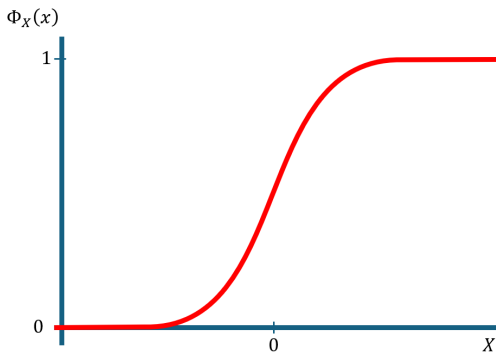


Fig. 4: The CDF of the Normal distribution, when $\mu = 0$ and $\sigma = 1$, which is the Standard Normal distribution



The PDF of the Normal distribution

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The general form of the PDF of the normal distribution is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

If it is the standard normal distribution, then $\mu = 0$ and $\sigma = 1$. Knowing that, the PDF of the standard normal distribution is:

$$\phi(x) = \frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}}$$



The PDF of the Standard Normal distribution

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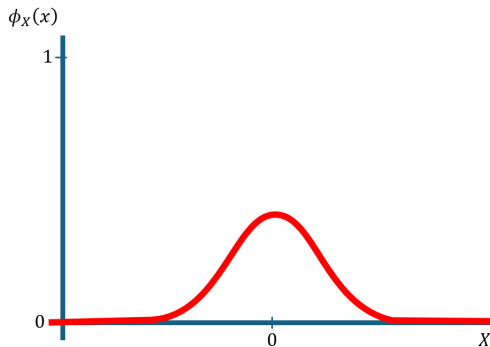


Fig. 5: The PDF of the Standard Normal distribution



The expectation and the variance of the Normal distribution

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The expectation is:

$$\mu$$

The median is:

$$\mu$$

The mode is:

$$\mu$$

The variance is:

$$\sigma^2$$



Different normal distributions and standardizing

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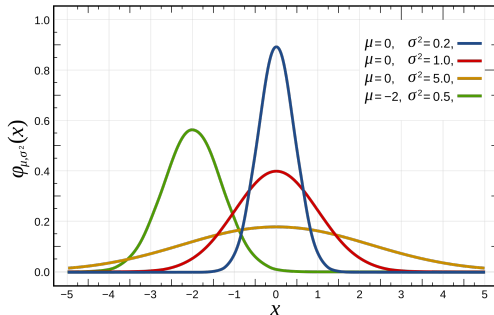


Fig. 6: Normal Distribution PDFs. Both μ and σ^2 are varied. Source: Wikipedia



Different normal distributions and standardizing

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Since technically every normal distribution is a transformation of the standard normal distribution, any point x from a normal distribution can be converted to the standard normal distribution z with the formula:

$$z = \frac{x - \mu}{\sigma}$$



Reason for standardizing

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STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.50000	.50399	.50798	.51197	.51595	.51994	.52392	.52790	.53188	.53586
0.1	.53983	.54380	.54776	.55172	.55567	.55962	.56356	.56749	.57142	.57535
0.2	.57926	.58317	.58706	.59095	.59483	.59871	.60257	.60642	.61026	.61409
0.3	.61791	.62172	.62552	.62930	.63307	.63683	.64058	.64431	.64803	.65173
0.4	.65542	.65910	.66276	.66640	.67003	.67364	.67724	.68082	.68439	.68793
0.5	.69146	.69497	.69847	.70194	.70540	.70884	.71226	.71566	.71904	.72240
0.6	.72575	.72907	.73237	.73565	.73891	.74215	.74537	.74857	.75175	.75490
0.7	.75804	.76115	.76424	.76730	.77035	.77337	.77637	.77935	.78230	.78524
0.8	.78814	.79103	.79389	.79673	.79955	.80234	.80511	.80785	.81057	.81327
0.9	.81594	.81859	.82121	.82381	.82639	.82894	.83147	.83398	.83646	.83891

Fig. 7: Table Values Represent AREA to the LEFT of the Z score.